

Data Dictionary: Mbala Municipal Council Web Dataset

CSC 4792 Data Mining and Warehousing — Mini Project
Project Team 17 — Mbala Municipal Council

1. Dataset Overview

Item	Description
Dataset name	db-unza26-csc4792-mbala_municipal_council_cleaned.csv
Council	Mbala Municipal Council
Project team	Project Team 17
Records	42 page-level records
Variables	8 columns
File format	CSV
Column separator	Pipe character ()
Source	Mbala Municipal Council official website
Primary use	Data mining, exploratory analysis, text analysis, information retrieval, and related municipal information analysis

2. Field-Level Data Dictionary

Field	Description	Data Type	Role / Notes
url	The URL of the original Mbala Municipal Council webpage from which the record was collected.	String	Source/provenance field; preserves the link to the originating webpage.
page_id	Identifier associated with the webpage record in the source dataset.	Numeric identifier	Used to identify and distinguish page-level records.
post_id	Identifier associated with the original post or content record.	Numeric identifier	Source identifier where available.
title	Title of the webpage, post, or publication represented by the record.	String	Provides a concise description of the page content.
date	Date associated with the webpage or publication.	Date / String	Stored in standardized YYYY-MM-DD form where a valid date was available.
topics	Topic or category labels associated with the webpage record.	String	May contain topic/category information used for grouping and analysis.
text_content	Textual content extracted from the corresponding webpage.	String	Main text field for text mining, NLP, search, and content analysis.
text_length_chars	Number of characters contained in the text_content field.	Integer	Derived field calculated from text_content; useful for descriptive text analysis.

3. Data Representation and Standardization

Aspect	Specification
Missing text values	Text fields were handled so that missing values do not prevent text-based processing.

Aspect	Specification
Whitespace	Text fields were standardized by removing unnecessary leading/trailing whitespace and normalizing repeated whitespace.
Date representation	Dates were standardized to YYYY-MM-DD where a valid date could be parsed.
Text length	text_length_chars is derived from the final text_content value.
CSV delimiter	The pipe character is used as the column separator.
Index	The exported CSV does not include the Pandas dataframe index.

4. Data Quality and Interpretation Notes

- The dataset contains 42 page-level records and 8 variables.
- The dataset represents information collected from the public digital footprint of Mbala Municipal Council.
- The URL field should be treated as the provenance reference for each webpage record.
- Identifier fields such as page_id and post_id originate from the source data and should not be interpreted as newly assigned analytical identifiers.
- text_length_chars is a derived variable and should be recalculated if text_content is modified.
- The dataset should be interpreted as a collection of webpage records rather than a complete representation of all activities, finances, projects, or services of Mbala Municipal Council.
- No unavailable figures or facts should be inferred from blank or incomplete source information.

5. Potential Analytical Uses

- Exploratory data analysis of municipal web publications.
- Topic-frequency and category analysis.
- Text-length and content-distribution analysis.
- Natural language processing and text mining.
- Information retrieval and search-oriented applications.
- Classification or clustering experiments using webpage metadata and text.
- Further integration with structured municipal project, CDF, LGEF, budget, or development records when such source data are available and properly documented.

Documentation note: This data dictionary describes the exact 8-column page-level dataset currently used in the Mbala Municipal Council CSC 4792 project. It is intended to accompany the cleaned CSV, Jupyter Notebook, README, and Data Description Paper.